34721/34722 Linear Control Design 1
Spring 2026

Teachers: Silvia Tolu - Dimitrios Papageorgiou

Lesson 3/13

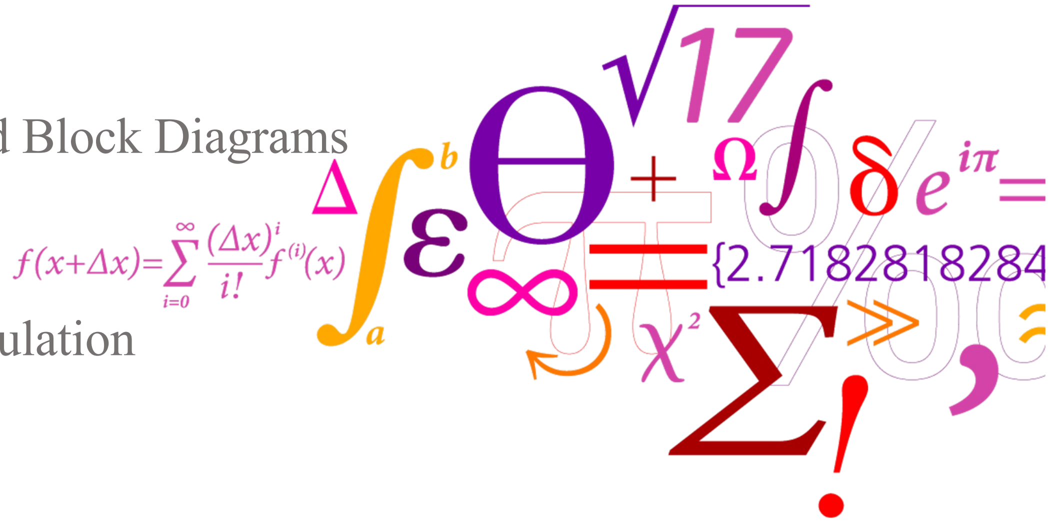
Schedule

13:00 - 15:00

- **Hand Tuning: PID**
- The Laplace Transformation
- Phasor
- Exercises: Transfer Functions and Block Diagrams

15:00 - 17:00

Exercises: Block Diagram Manipulation

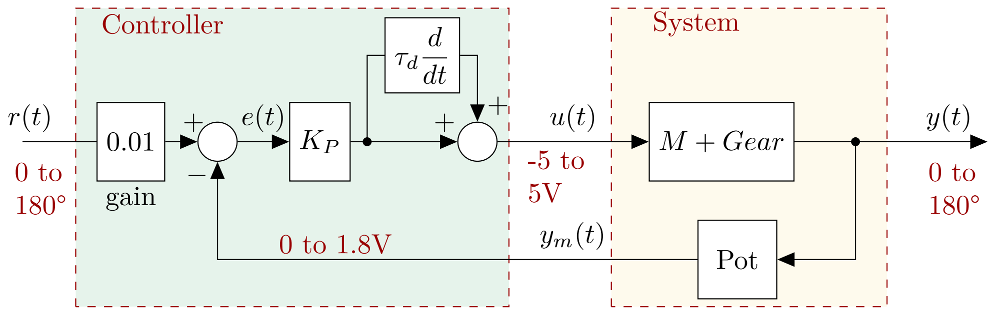


DTU ELECTRO
Department of Electrical and Photonics Engineering
Control, Robotics and Embodied AI (CREA) Group

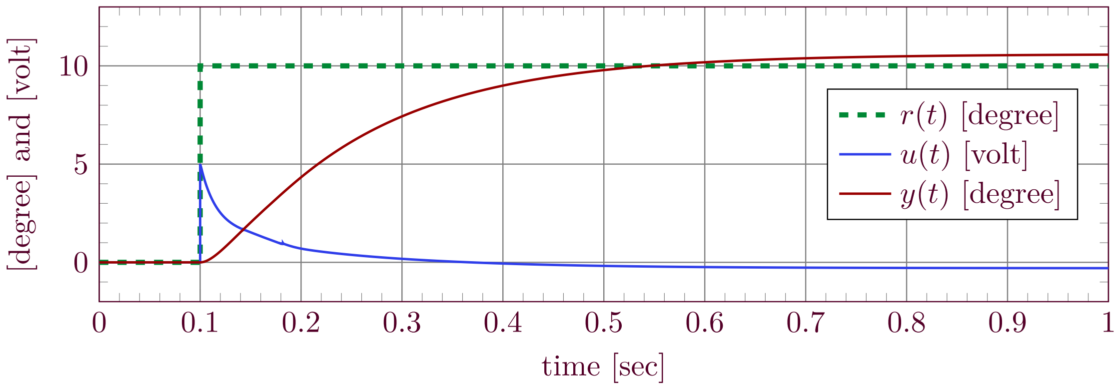
Day	Topics	Contents	Assignments
1	Introduction	Intro to feedback control, Pre-Test	MATLAB and Simulink
2	Control concepts	Block diagram, control process, hand tuning	Hand tuning, Ziegler-Nichols
3	**Transfer function and Laplace**	**Laplace and transfer function**	**Block diagrams and transfer functions**
4	Frequency domain	Frequency domain properties	Introduction to REGBOT
5	Modelling and Linearization	White/black box modelling and linearization	REGBOT – Transfer function from data
6	Bode Plots and stability	Bode plot, stability margins, P-design	REGBOT – Bode, stability, P-Controller design

PD-Controller

$$u(t) = K_p\,(\,e(t) + \tau_d \frac{d}{dt} e(t)\,)$$



$$K_P = 50$$
$$\tau_d = 0.1\,\mathrm{s}$$

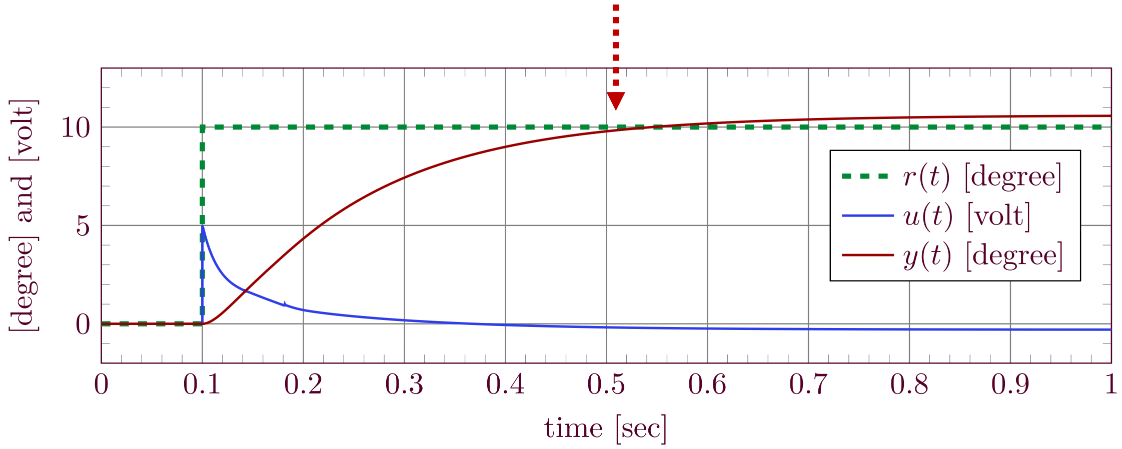


RECIPE

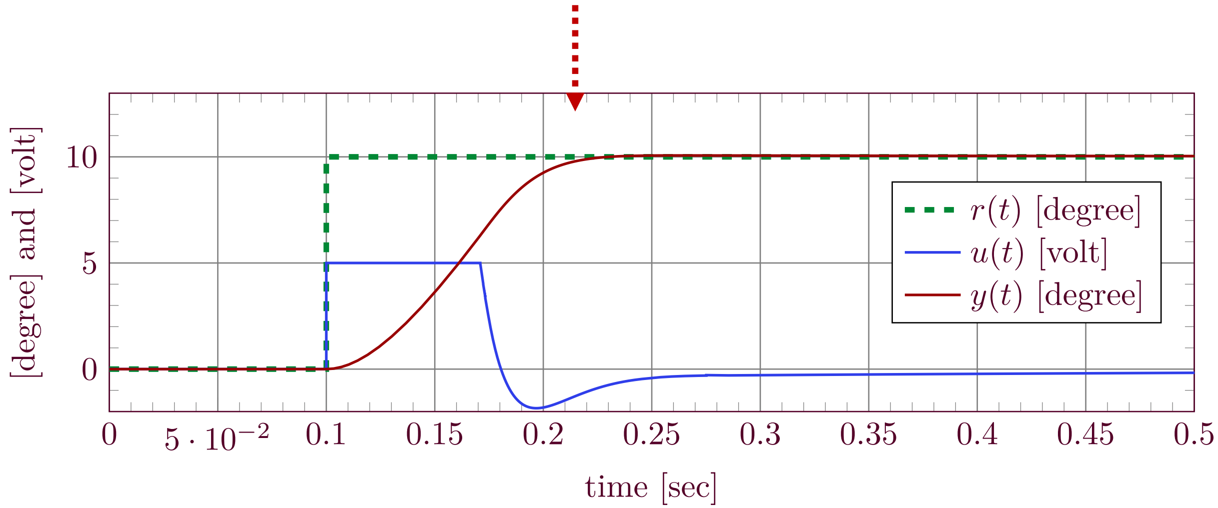
When there is a fast change in the reference input (step), the derivative term quickly increases the control signal u(t).

- Start with a reasonable K_p
- Look at the rise time and select a τ_d in the same order of magnitude.
- If there is an oscillation or too much overshoot ⇒ increase τ_d and decrease K_p
- To make the system faster ⇒ decrease τ_d and increase K_p

PD-Controller Refinement



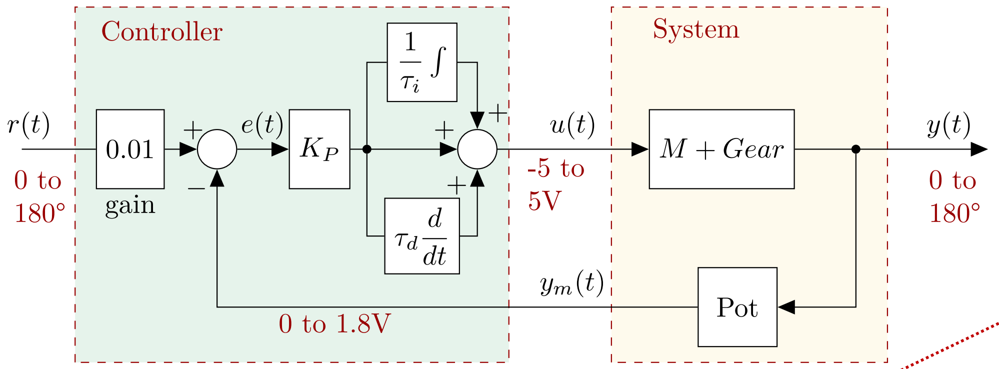
$$K_P = 50$$
$$\tau_d = 0.1\,\mathrm{s}$$



$$K_P = 500$$
$$\tau_d = 0.02\,\mathrm{s}$$

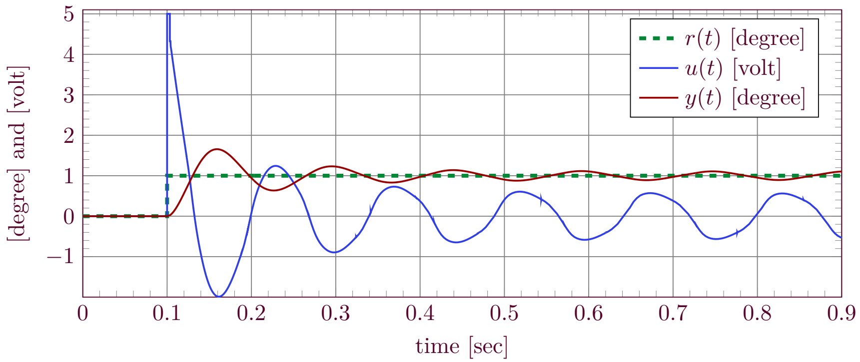
PID-Controller

$$u(t) = K_p \left(e(t) + \frac{1}{\tau_i}\int e(t)dt + \tau_d \frac{d}{dt}e(t)\right)$$

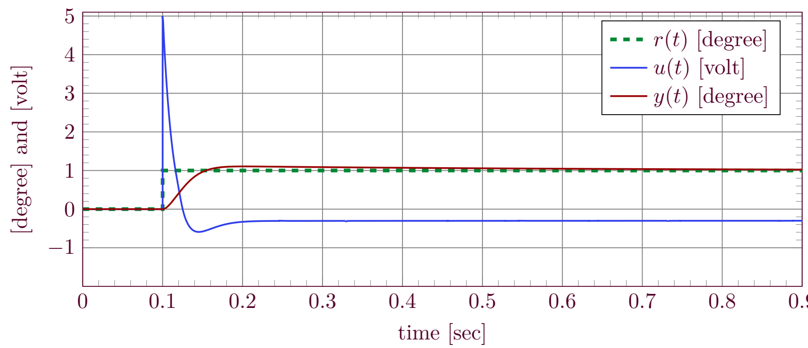


the I-term is just too fast for this example. When the I-term is slower, then the static error is removed slower.

$$K_P = 500$$
$$\tau_i = 0.02\,\mathrm{s}$$
$$\tau_d = 0.02\,\mathrm{s}$$



$$K_P = 500$$
$$\tau_i = 0.5\,\mathrm{s}$$
$$\tau_d = 0.02\,\mathrm{s}$$



Hand Tuning of PID Parameters

Initial parameter estimates

Proportional gain ($\boldsymbol{K_p}$)

Start with an estimate based on a reasonable operating point change: $\boldsymbol{K_p \approx \frac{u_0}{e_0}}$

e_0: expected initial error (e.g., from a step change)
u_0: estimated control effort required to correct that error
Choose values that produce a noticeable but not excessive system response.

Integral time ($\boldsymbol{\tau_i}$)
A reasonable initial value is equal to the time for the output to reach steady state

Derivative time ($\boldsymbol{\tau_d}$)
A practical starting point is $\tau_d \approx \tau_i$ or based on the observed rise time of the step response.

Hand Tuning of PID Parameters (Refinement)

Proportional Gain (Kp)

Increase Kp

→ Faster response
→ Reduced steady-state error
→ Higher risk of overshoot or instability

Decrease Kp

→ Improved stability
→ Less overshoot and oscillations
→ Slower response

Integral Time (τ_i)

Decrease τ_i (stronger integral action)
→ Faster elimination of steady-state error
→ More risk of oscillation

Increase τ_i (weaker integral action)
→ More stable response
→ Slower correction of steady-state error

Derivative Time (τ_d)

Decrease τd
→ Faster rise time
→ Less damping

Increase τd
→ More damping
→ Less overshoot
→ Slower overall response

Modelling and applications of models

Physical system

Electromechanical model

Mathematical model

Assumptions
(simplifications)

Newton, Euler, D'Alembert,
Lagrange, Hamilton …
Coulomb, Ampère, Ohm,
Kirchhoff, Faraday, Lorentz …
(First principles)

Non-linear systems $\dot{x}(t) = f(x(t), u(t), t)$

$y(t) = g(x(t), u(t), t)$

Stationary point **Linearization + Laplace Transformation**

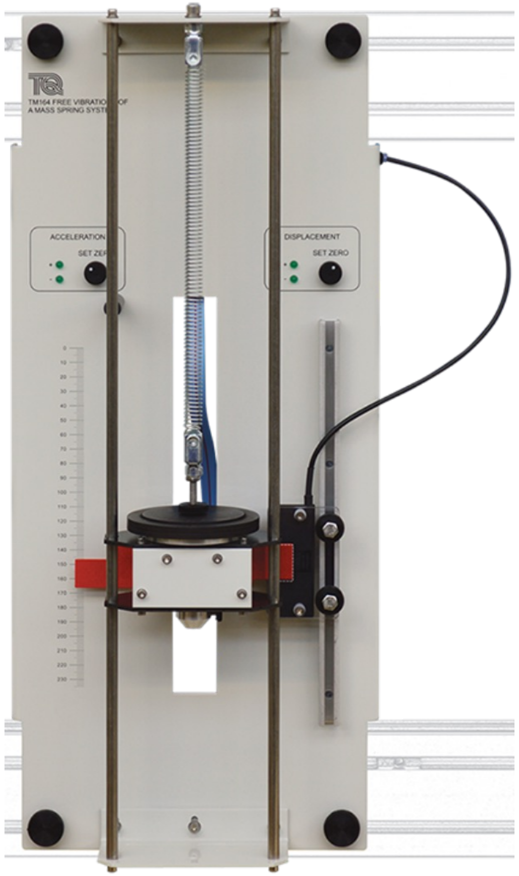
Transfer function $\boldsymbol{Y(s) = G(s)\, U(s)}$

Physical system

Assumptions?

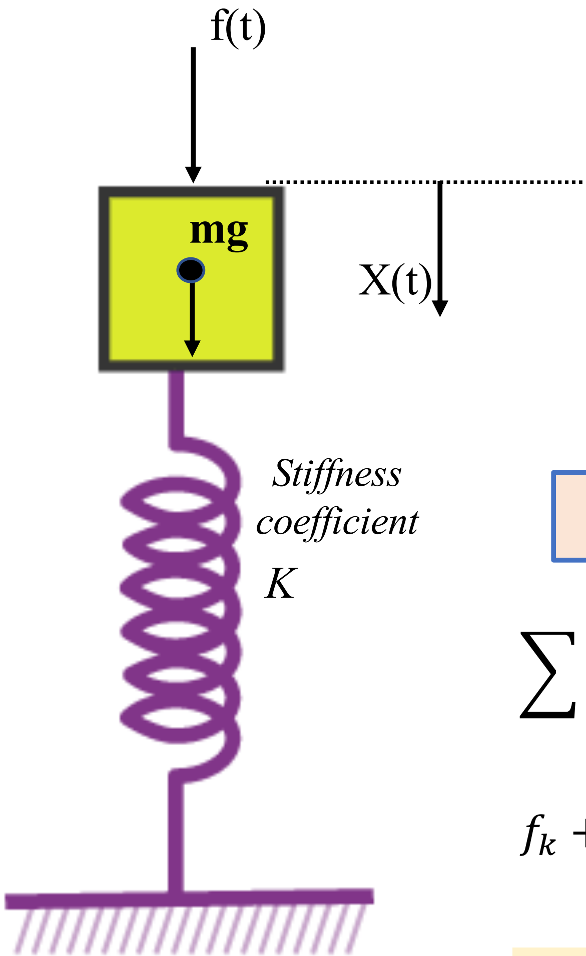
- The object is a particle

- No distribution of mass around axes, no inertia



https://www.tecquipment.com/free-vibration-of-a-mass-spring-system

Mechanical model



f(t) Perturbation (i.e., force I apply with a finger)

g gravity

$-K \cdot x(t)$ Spring force

What do we want to measure??
X(t) displacement

Mathematical model

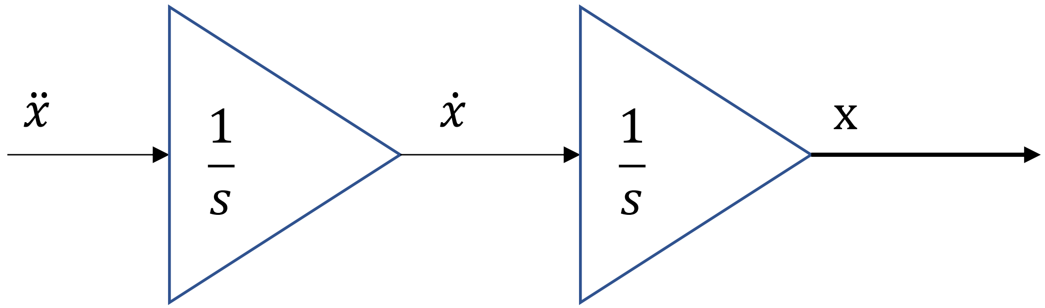
$$\sum F = m\,\ddot{x}(t) \qquad \text{Newton's Law}$$

$$f_k + f(t) + mg = m\ddot{x}(t)$$

$$\ddot{x}(t) = -\frac{k}{m}x(t) + g + \frac{1}{m}f(t)$$

Block Diagram

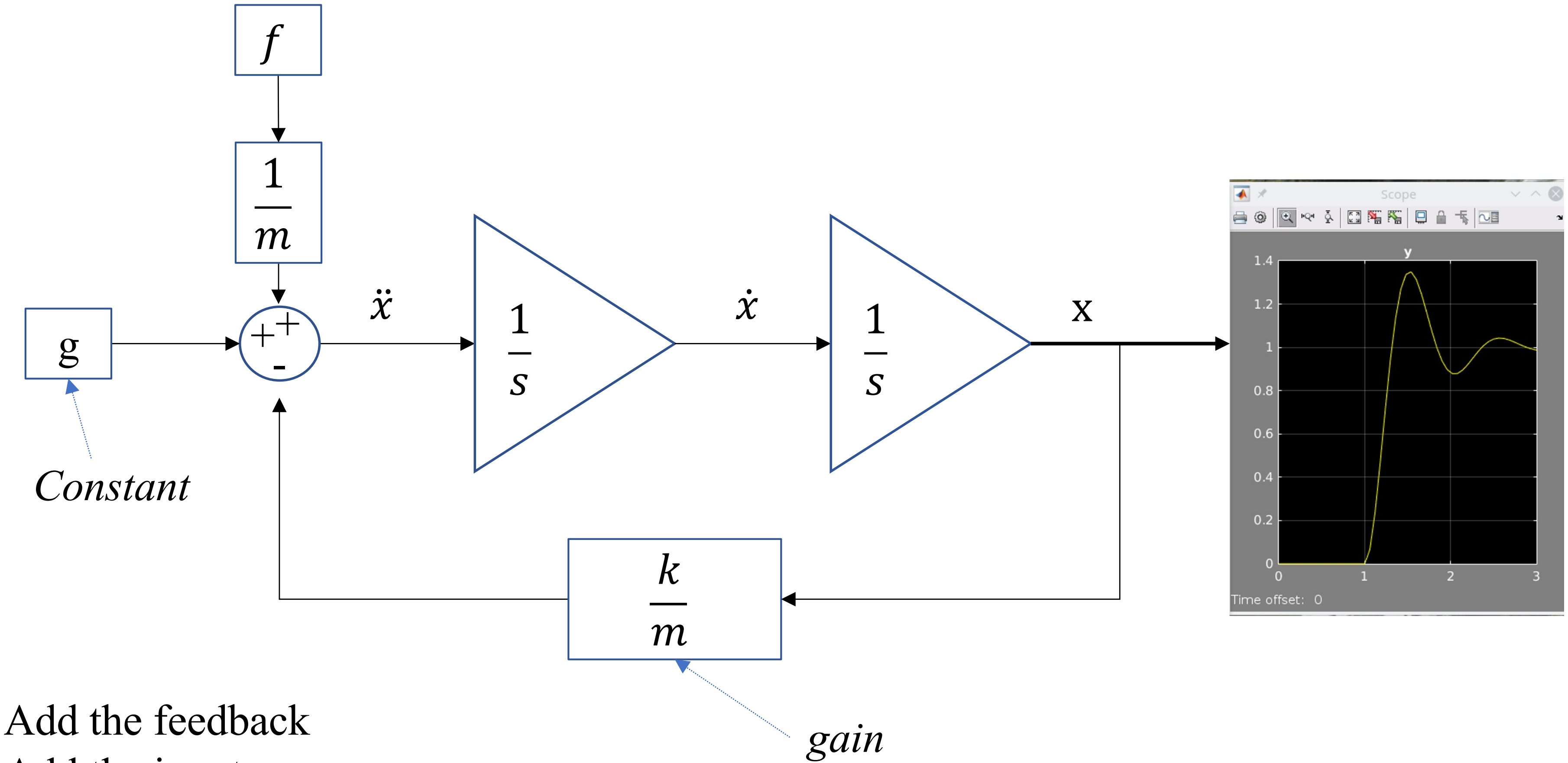
$$\ddot{x}(t) = -\frac{k}{m}x(t) + g + \frac{1}{m}f(t)$$



1. We isolate the term with the highest order (i.e., ***acceleration***)
2. Add integrators ($\frac{1}{s}$) as many as the order of the equation

Block Diagram

$$\ddot{x}(t) = -\frac{k}{m}x(t) + g + \frac{1}{m}f(t)$$



3. Add the feedback
4. Add the input
5. Set constants and gains (g, m, k)

Frequency - Laplace

The Laplace Transform allows the transformation from the time domain ***f(t)*** to and from the complex frequency domain.

$$s = \sigma + j\omega$$

$j\boldsymbol{\omega}$ is the complex part of **s** and it is the same as the physical frequency.

➔ We can interpret the **frequency response** in the Laplace **[s]** domain by looking at the part of the complex frequency domain $\boldsymbol{\sigma} = 0$, $s = j\boldsymbol{\omega}$

The Laplace Transform

$$F(s) = \mathscr{L}\{f(t)\} = \int_0^{\infty} f(t)e^{-st}dt$$

Properties

$f(t)$= 0 for $t < 0$ (a real argument t)

A differential equation becomes a **polynomial** (simple arithmetic rules)

s Can be interpreted as a frequency $s = \sigma + j\omega$

Inverse Laplace Transformation

$$\mathscr{L}{-1}\{F(s)\} = f(t) = \frac{1}{2\pi j}\int_{\sigma_1 - j\infty}^{\sigma_1 + j\infty} F(s)e^{st}dt$$

In electrical terms:
j denotes the imaginary part of a number (since ***i*** denotes current).

Laplace Calculation Rules

$f(t)$	$\mathbf{F}(s)$	Operation
$Af(t)$	$A\mathbf{F}(s)$	Scaling
$f_1(t) + f_2(t)$	$\mathbf{F}_1(s) + \mathbf{F}_2(s)$	Addition
$f_1(t) - f_2(t)$	$\mathbf{F}_1(s) - \mathbf{F}_2(s)$	Subtraction
$\frac{d}{dt}f(t)$	$s\mathbf{F}(s) - f(0)$	Differential
$\frac{d^2}{dt^2}f(t)$	$s^2\mathbf{F}(s) - sf(0) - f'(0)$	Differential
$\int_0^t f(\tau)d\tau$	$\frac{1}{s}\mathbf{F}(s)$	Integrate
$\lim_{t\to 0} f(t)$	$\lim_{s\to\infty} s\mathbf{F}(s)$	Start value
$\lim_{t\to\infty} f(t)$	$\lim_{s\to 0} s\mathbf{F}(s)$	End value

Laplace and Signals

Example:

$$f(t) \rightarrow \boxed{} \rightarrow f_2(t)$$

$$f(t) = 2$$

$$f_2(t) = \frac{1}{m}\int f(t)dt$$

Laplace domain

$$F(s) = 2\frac{1}{s}$$

$$F_2(s) = \frac{1}{m}\frac{1}{s}F(s)$$

$$F_2(s) = \frac{2}{m}\,\frac{1}{s^2}$$

Time domain

$$f_2(t) = \frac{2}{m}t$$

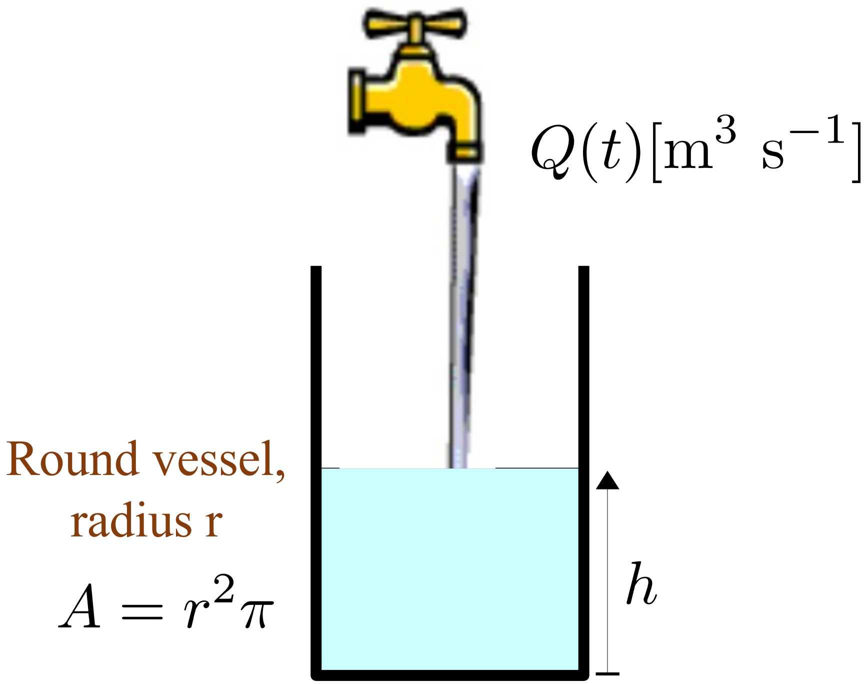
$f(t)$	$\mathbf{F}(s)$	Description of *f(t)*
$\delta(t)$	1	unit impulse
$u(t)$	$\frac{1}{s}$	unit step
e^{-at}	$\frac{1}{s+a}$	[1]) exponential
t	$\frac{1}{s^2}$	
t^2	$\frac{2}{s^3}$	
te^{-at}	$\frac{1}{(s+a)^2}$	[1])
$\sin(bt)$	$\frac{b}{s^2+b^2}$	Sinus
$\cos(bt)$	$\frac{s}{s^2+b^2}$	Cosinus
$e^{-at}\sin(bt)$	$\frac{b}{(s+a)^2+b^2}$	[1])

[1]) a > 0 for curve

Transfer function

h as a function of Q

h(t) = ?



Volumetric flow rate
Q = h A= h r$^2\pi$ = h A

Laplace

H(s) = ?

Transfer function

$$\frac{H(s)}{Q(s)} = ?$$

Block Diagram

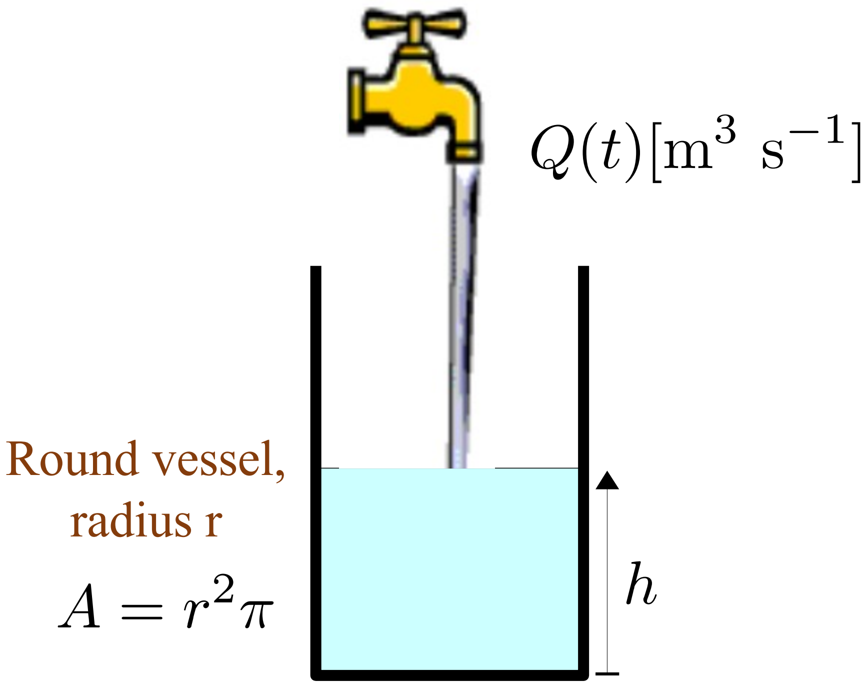
?

Transfer function

h as a function of Q

Volumetric flow rate

$Q = \dot{h}\, r^2\pi = \dot{h}\, A$



$$h(t) = \frac{1}{A}\int_0^t Q(t)dt$$

$$h(t) = \frac{1}{r^2\pi}\int_0^t Q(t)dt$$

Laplace

$$H(s) = \frac{1}{r^2\pi}\frac{1}{s}Q(s)$$

Transfer function

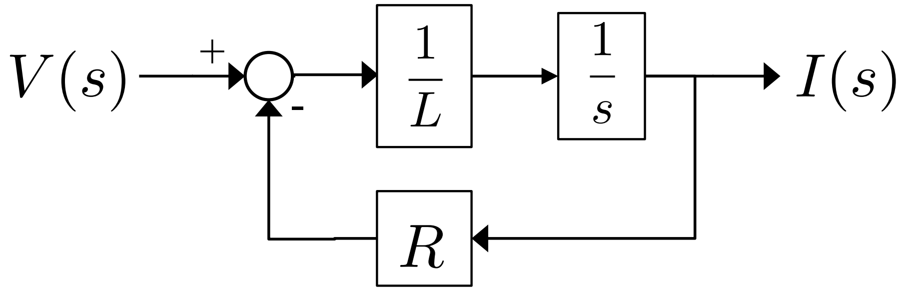
$$\frac{H(s)}{Q(s)} = \frac{1}{r^2\pi\, s}$$

Block Diagram

$$Q(s) \rightarrow \boxed{\frac{1}{r^2\pi}} \xrightarrow{\dot{H}} \boxed{\frac{1}{s}} \rightarrow H(s)$$

Example: Transfer function and steady state value

The relationship of input voltage **V(s)** and output current **I(s)** for a circuit is modelled as follows:



a) What will be the **transfer function** if L is made smaller?

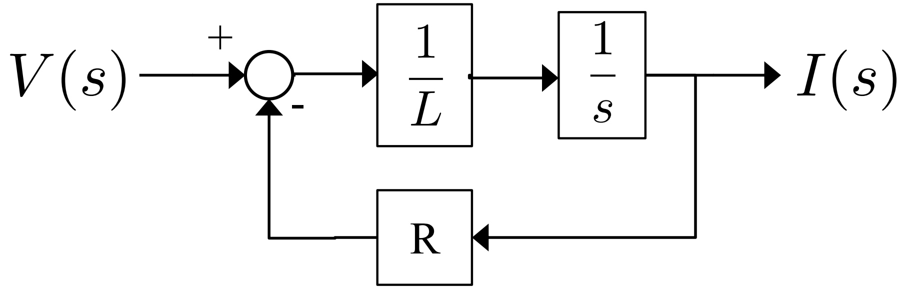
$$\lim_{L \to 0} \frac{I(s)}{V(s)} =$$

b) What will be the **steady-state** value for a step input (A Volts)?

$$I_{ss} = \lim_{t \to \infty} I(t) =$$

Example: Transfer function

a) **What will be the transfer function if L is made smaller:**



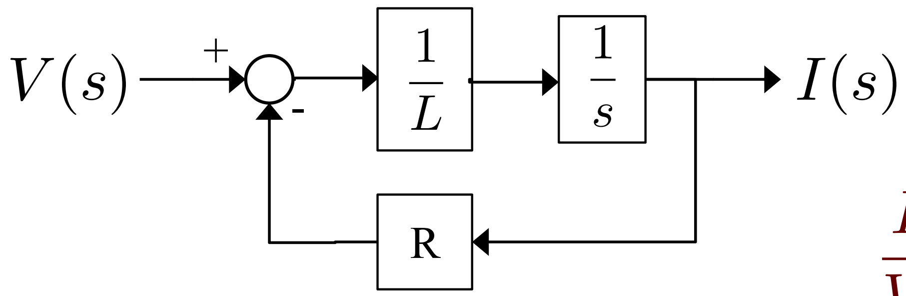
$$\frac{I(s)}{V(s)} = \frac{1}{Ls(1 + \frac{1}{Ls}R)} = \frac{1}{Ls + R}$$

$$\lim_{L \to 0} \frac{I(s)}{V(s)} = \lim_{L \to 0} \frac{1}{Ls + R} = \frac{1}{R}$$

Example: steady state value

b) What will be the steady-state value for a step input (A Volt)?

$$V(s) = \frac{A}{s}$$



$$\frac{I(s)}{V(s)} = \frac{1}{Ls+R}$$

$$I_{ss} = \lim_{t\to\infty} I(t) = \lim_{s\to 0} s\frac{A}{s}\frac{1}{Ls+R}$$

$$I_{ss} = \frac{A}{R}$$

$\lim_{t\to 0} f(t)$	$\lim_{s\to\infty} s\mathbf{F}(s)$	Start value
$\lim_{t\to\infty} f(t)$	$\lim_{s\to 0} s\mathbf{F}(s)$	End value

Questions

1) A model consists of a block with the transfer function **G(s)**, a constant input voltage **v(t) = 5** and the output is a speed $\dot{x}(t)$

a) How is the voltage expressed in the Laplace domain $V(s) =$?

b) What is $\dot{x}(s)$, if $G(s) = \dfrac{s}{s+10}$?

c) What is $\dot{x}(t)$?

2) If $v(t) = e^{-3t}$, what is $\dot{x}(t)$?

Control Question - Answer (1.a)

1) A model consists of a block with the transfer function **G(s)**, a constant voltage input **v(t) = 5** and the output is a speed $\dot{x}(t)$

a) How is the voltage expressed in the Laplace domain $V(s) = \ ?$

Since v(t) = 0 for t<0, the voltage is a step: $\boldsymbol{V(s) = \frac{5}{s}}$

Control Question - Answer (1.b)

1) A model consists of a block with the transfer function **G(s)**, a constant voltage input **v(t) = 5** and the output is a speed $\dot{x}(t)$

a) $V(s) = \frac{5}{s}$

b) What is $\dot{x}(s)$**, if** $G(s) = \frac{s}{s+10}$?

$$\dot{x}(s) = V(s)G(s) = \frac{5}{s}\frac{s}{s+10}$$

$$\dot{x}(s) = \frac{5}{s+10}$$

Control Question - Answer (1.c)

1) A model consists of a block with the transfer function **G(s)**, a constant voltage input **v(t) = 5** and the output is a speed $\dot{x}(t)$

a) $$V(s) = \frac{5}{s}$$

b) $$\dot{x}(s) = V(s)G(s) = \frac{5}{s}\frac{s}{s+10}$$

$$\dot{x}(s) = \frac{5}{s+10}$$

c) What is $\dot{x}(t)$ **?** $$\dot{x}(t) = 5e^{-10t}$$

e^{-at}	$\frac{1}{s+a}$

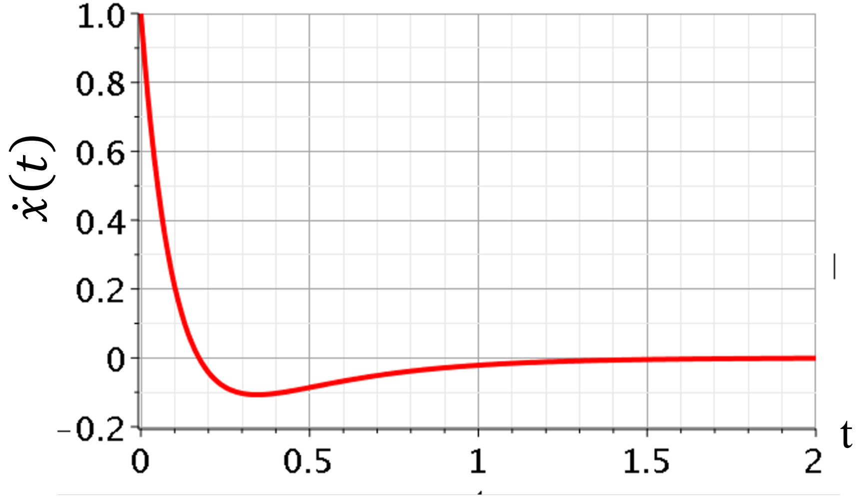
Control Question - Answer (2)

2) If $v(t) = e^{-3t}$, what is $\dot{x}(t)$?

Matlab

```
syms a s t w x
Vs = laplace(exp(-3*t))
Gs = s/(s+10) % Transfer Function
Xds = Vs * Gs % output

Xdt = ilaplace(Xds)
t=0:0.01:2
Xdtt = eval(Xdt)
plot(t, Xdtt)
```



$$v(t) = e^{-3t} \Rightarrow V(s) = \frac{1}{s+3}$$

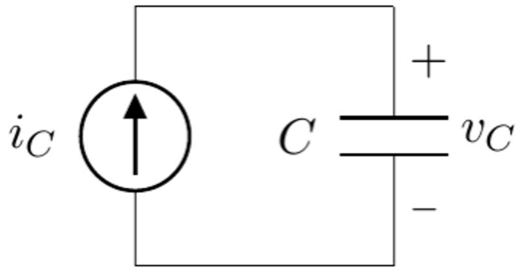
$$\dot{X}(s) = V(s)G(s) \implies \dot{X}(s) = \frac{1}{s+3}\frac{s}{s+10}$$

Use fraction expansion operation or Matlab

$$\dot{X}(t) = 1.43e^{-10t} - 0.43e^{-3t)}$$

Electronic Components and Laplace

For a capacitor with a capacitance **C** (in Farad) the relationship between voltage and current (time domain) is as follows:



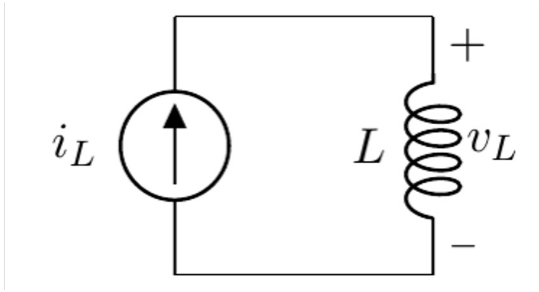
$$v_C(t) = \frac{1}{C}\int_0^t i_C(\tau)d\tau$$

- **Calculate the expression** $V_c(s)$ $\quad V_C(s) = \frac{1}{sC} I_C(s)$

- **What is the transfer function from current to voltage?** $\quad Z(s) = \frac{V_C(s)}{I_C(s)} = \frac{1}{sC}$

Electronic Components and Laplace

For a coil with the inductance **L** (in Henry), the relationship between voltage and current in the Laplace domain is as follows:



$$V_L(s) = sLI(s)$$

- **What is the expression of** $v_L(t)$ **in the time domain?** $v_L(t) = L\frac{d}{dt}i(t)$

- **What is the transfer function from current to voltage?** $Z_L(s) = \frac{V_L(s)}{I(s)} = sL$

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Lesson 3/13

Schedule

13:00 - 15:00

- Hand Tuning: PD
- The Laplace Transformation
- ➢ Phasor
- Exercises: Transfer Functions and Block Diagrams

15:00 - 17:00

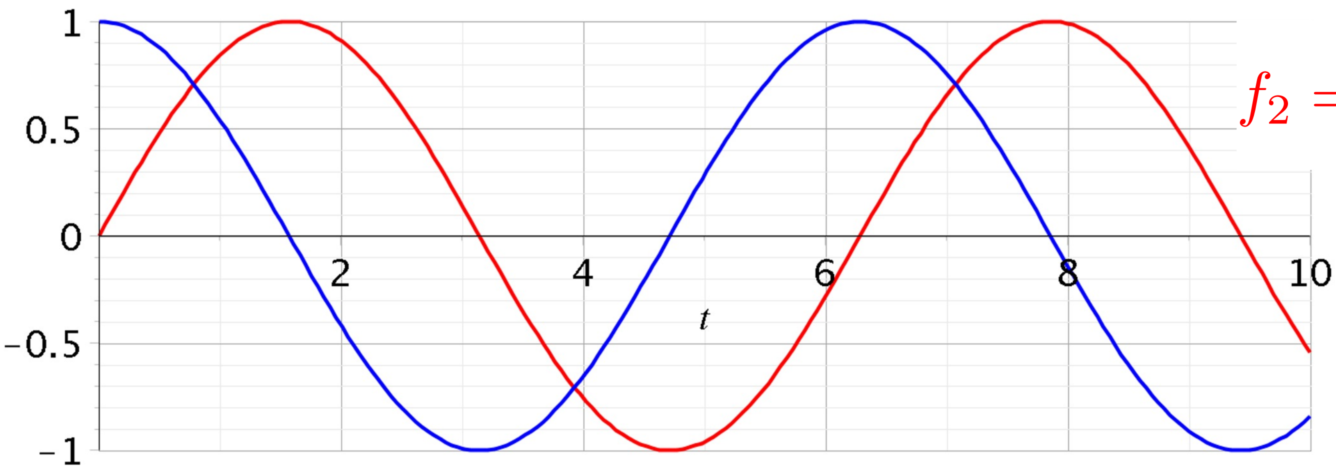
Exercises: Block Diagram Manipulation

Frequency

In a linear system, if a sinusoidal signal is ***in****, then a sinusoidal signal is* ***out*** *with the same frequency, but the phase and amplitude may have changed.*

$$f_1(t) \rightarrow \boxed{k\int_0^t} \rightarrow f_2(t)$$

$$f_1(t) = cos(\omega t) \qquad \omega = 1$$



$$f_2 = \mathrm{k}\int_0^t f_1 \qquad \mathrm{k} = 1$$

ω [rad s^{-1}] angular frequency

$\omega = 2\pi f$

$f = \dfrac{\omega}{2\pi}$ [Hz]

MATLAB

```
dt = 0.001;
t = 0:dt:10;
k = 2;
f = 1/(2*pi);
w = 2*pi*f;
f1 = k * cos(w*t);
f2 = cumsum(f1*dt);
figure(1), hold on;
plot(t, f1,'b');
plot(t, f2,'r');
grid on
hold off;
```

What is a phasor?

$f(\omega, t) = A\cos(\omega t + \theta)$

A amplitude
θ phase
ω frequency

It is a complex number with A and θ

$F = Ae^{j\theta} = A\,(\cos(\theta) + j\sin(\theta))$ *Euler notation*

or

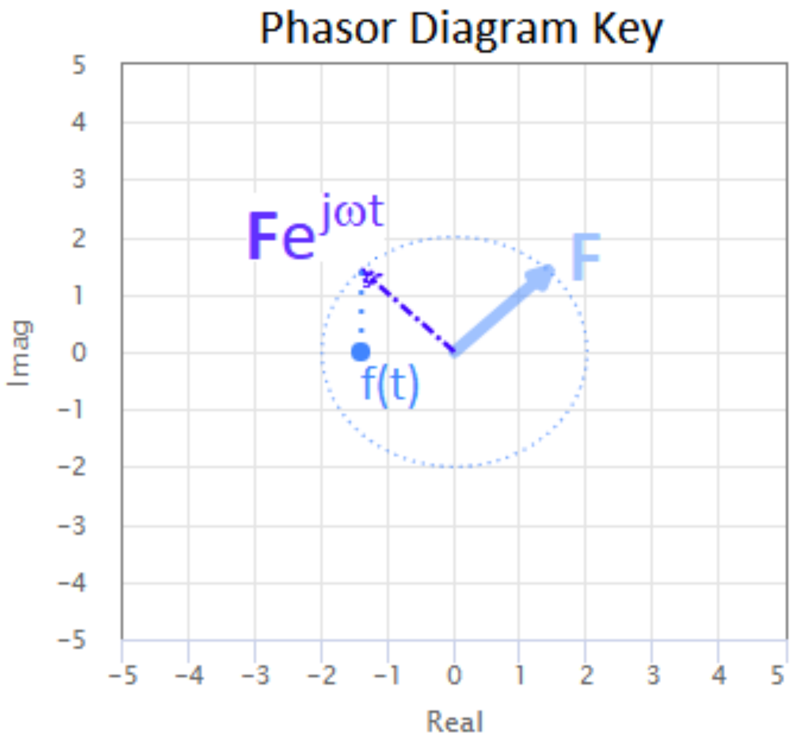
$\mathrm{F} = \mathrm{A}\angle\theta$

We rotate F by an angle ωt

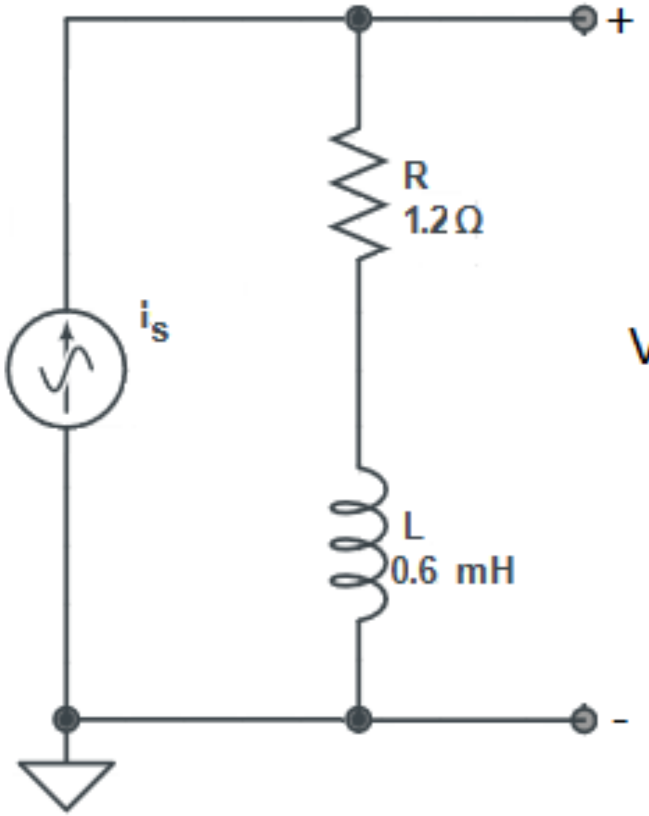
$\mathrm{F}e^{j\omega t} = Ae^{j\theta}e^{j\omega t} = Ae^{j(\omega t+\theta)} = \mathrm{A}\angle(\mathrm{wt} + \theta)$

Back to time domain

$\mathrm{f(t)} = \mathrm{Re}\{\mathrm{F}e^{j\omega t}\}$



Why a phasor?



$i_s(t) = 2\cdot\cos(1000\cdot t)$
$\omega = 1000$ rad/sec

$\rightarrow$ $\mathbf{I_s} = 2\angle 0$

$V_R(s) = R\cdot\mathbf{I_s}$

$V_L(s) = j\omega L\cdot\mathbf{I_s}$

$V_{out}(s) = (R + j\omega L)\cdot\mathbf{I_s}$

$\mathbf{Z} = R + j\omega L$ is a complex number

$\mathbf{Z} = 1.2 + j\,0.6 = M\angle\phi$

$M = \sqrt{1.2^2 + 0.6^2} = 1.34$

$\Phi = \text{atan2}(\omega L, R) = \text{atan2}(6, 1.2) = 26.6^o$

$\mathbf{Z} = 1.34\angle 26.6^o$

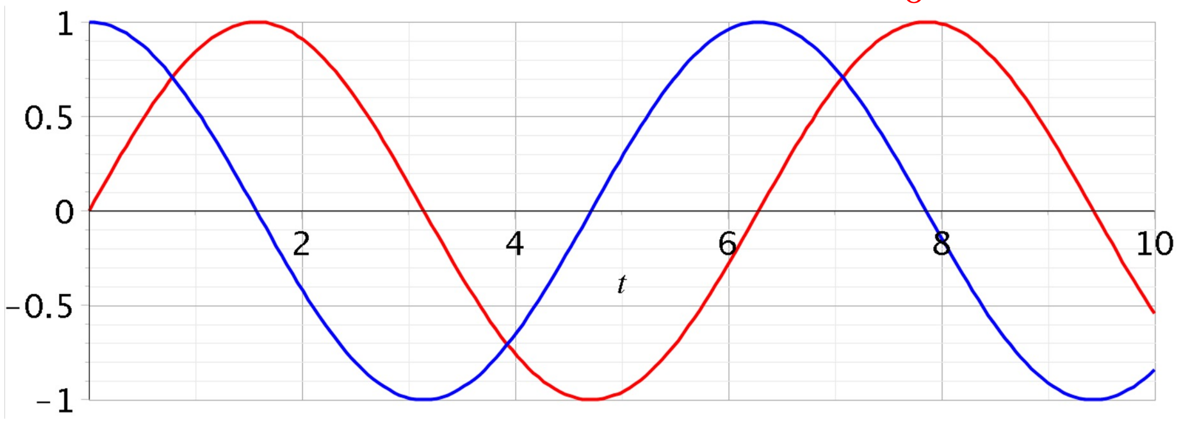
$V_{out}(s) = \mathbf{Z}\,\mathbf{I_s} = (2\angle 0)\cdot(1.34\angle 26.6°) = 2.68\angle 26.6°$ ← **the phasor form of Ohm's law**

$\mathbf{V_{out}(t)} = 2.68\cdot\cos(1000\cdot t + 26.6°)$

We use only algebra!

Frequency – Laplace / Phasor

$f_1(t) = cos(\omega t)$ $\omega = 1$ $f_2 = k\int_0^t f_1$ $k = 1$



Laplace expression can be interpreted as Phasor, when

$$s = \sigma + j\omega \,\Big|\, \begin{matrix}\sigma = 0, \\ \omega > 0\end{matrix}$$

$$\Rightarrow s = j\omega$$

Amplitude and phase shift at a frequency ω can be derived directly in the Laplace domain.

$$f_2(t) = k\int_0^t f_1(t)\ dt$$

$$f_1(t) \longrightarrow \boxed{k\int_0^t} \longrightarrow f_2(t)$$

Laplace

$$F_1(s) \longrightarrow \boxed{\frac{k}{s}} \longrightarrow F_2(s)$$

$$F_2(s) = \frac{k}{s}\ F_1(s)$$

Phasor

$$F_2(j\omega) = \frac{k}{j\omega}\ F_1(j\omega)$$

Phasor is subset of Laplace

Laplace and Phasor - Example I

A circuit with the transfer function

$$G(s) = \frac{x(s)}{u(s)} = \frac{10}{s + 312}$$

What is the transfer function with gain M **and phase shift** φ

at the frequency $\omega = 628 \text{ rad/s}$ **and** $u(t) = cos(\omega t)$ **?**

Example I – Answer

$$G(j\omega) = \frac{x(j\omega)}{u(j\omega)} = Me^{j\varphi}$$

$$G(j\omega) = \frac{x(j\omega)}{u(j\omega)} = \frac{10}{j\omega + 312} \qquad G(M,\varphi) = \frac{10\, e^{j0}}{Ae^{j\phi}}$$

$$A = \sqrt{\omega^2 + 312^2}, \qquad \phi = arctan\frac{\omega}{312}$$

$$M = \frac{10}{A} = 0.0413 \qquad \varphi = 0 - \phi = -1.11 \text{ rad} = -63.6^o$$

$$G(j\omega) = 0.0143\, e^{-j63.6}$$

Example I - DO it in MATLAB

$$G(s) = \frac{x(s)}{u(s)} = \frac{10}{s + 312}$$

% define s variable
% define transfer function
% Evaluate at s = omega*j - help *evalfr NB j is i in Matlab*
% find magnitude - help *abs*
% find phase in degrees - help *angle*

Example I - Do it in MATLAB

$$G(s) = \frac{x(s)}{u(s)} = \frac{10}{s + 312}$$

```
s = tf('s’); % define s variable
G = 10/(s + 312); % define transfer function
z = evalfr(G, 628*i); % Evaluate at s = omega*j
z_mag = abs(z); % find magnitude
z_phase = rad2deg(angle(z)); % find phase in degrees
```

M = 0.0143
φ = - 63.6°

Mass-Spring-Damper System

$$M\ddot{y} = -ky - \beta\dot{y} + F$$

$$G(s) = \frac{\omega_0^2}{s^2 + 2\zeta\omega_0 s + \omega_0^2}$$

- Spring constant $\boldsymbol{k = \omega_0^2}$
- Damping coefficient $\boldsymbol{\beta = 2\zeta\omega_0}$ **(damping factor $\boldsymbol{\zeta}$)**
- Mass M (e.g., M = 1)

$$y(s) = \frac{40}{s^2 + 4s + 40}u(s) \Leftrightarrow \ddot{y} = -40y - 4\dot{y} + 40u$$

Laplace and Phasor - Example II

A system has the following transfer function (**Second-Order System**):

$$u(s) \longrightarrow \boxed{\frac{40}{s^2 + 4s + 40}} \longrightarrow y(s)$$

a) If u is a cosine with the amplitude equal to 1 $u(t) = \cos(\omega t)$

Find the amplitude and phase shift of $y(t)$ **when:**

$$\omega = 1$$

$$\omega = 1000$$

$$\omega = \sqrt{40}$$

Example II

A system has the following transfer function:

$$u(s) \longrightarrow \boxed{\frac{40}{s^2+4s+40}} \longrightarrow y(s)$$

a) If u is a cosine with the amplitude 1: $u(t) = \cos(\omega t)$

Find the amplitude and phase shift of $y(t)$ **when:**

$$G(\omega) = \frac{40}{\underbrace{(j\omega)^2}_{-1\cdot\omega^2} + 4j\omega + 40}$$

real

Group real and imaginary terms

Example II – Answer a)

$$G(\omega) = \frac{40}{(j\omega)^2 + 4j\omega + 40}$$

$$G(\omega) = \frac{40}{(-\omega^2 + 40) + 4j\omega} \Rightarrow |G(s)| = \left| \frac{40}{(-\omega^2 + 40) + 4j\omega} \right| = \frac{|40|}{|(-\omega^2 + 40) + 4j\omega|}$$

$$= \frac{40}{\sqrt{(-\omega^2 + 40)^2 + 4^2\omega^2}}$$

$\alpha \quad + \ \beta j$

$M = \frac{40}{A_d} \quad A_d = \sqrt{(-\omega^2 + 40)^2 + 4\omega^2}$

$\phi = \arctan(4\omega/(-\omega^2 + 40))$

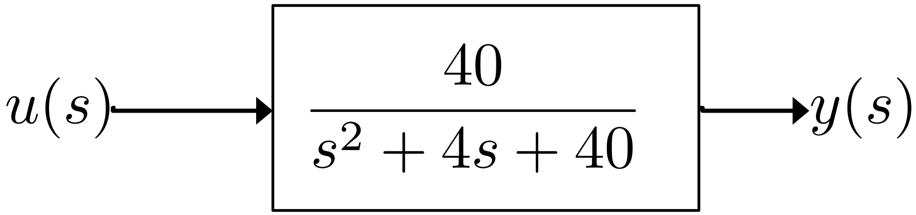
$\varphi = 0 - \phi$

$\omega = 0.1 \frac{rad}{s} \Longrightarrow G(\omega) \approx 1.0 \angle - 0.57^\circ$

$\omega = 1000 \frac{rad}{s} \Longrightarrow G(\omega) \approx 40 \cdot 10^{-6} \angle - 180^\circ$

$\omega = \sqrt{40} \frac{rad}{s} \Longrightarrow G(\omega) \approx 1.5 \angle - 90^\circ$

Example II - Do it in MATLAB



% define s variable
% define transfer function
% Evaluate at s = omega*j, e.g., omega = 0.1
% find magnitude
% find phase in degrees

Example II - Do it in MATLAB

$$u(s) \longrightarrow \boxed{\frac{40}{s^2 + 4s + 40}} \longrightarrow y(s)$$

```
s = tf('s') % define s variable
G = 40 / (s^2 + 4*s + 40) % define transfer function
z = evalfr(G, 0.1*i) % Evaluate at s = omega*j, e.g., omega = 0.1
z_mag = abs(z) % find magnitude
z_phase = rad2deg(angle(z)) % find phase in degrees
```

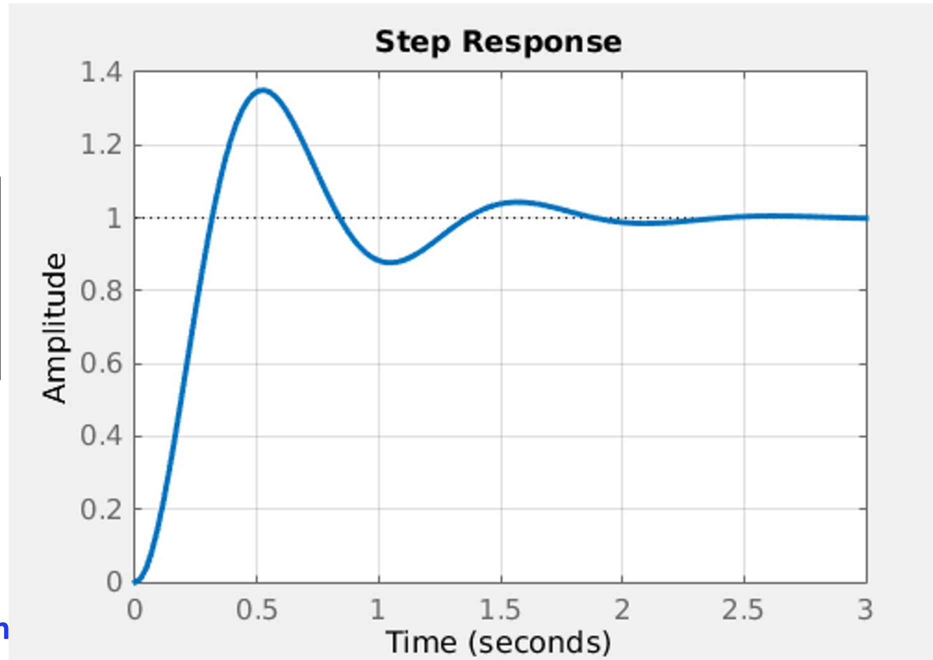
Example II – b) Step input - MATLAB

A system has the following transfer function:

$$u(s) \longrightarrow \boxed{\frac{40}{s^2 + 4s + 40}} \longrightarrow y(s)$$

With a step of magnitude 1, what is the output as a function of time?

```
% define s variable
% define transfer function
% Time response to a step of 1 (from 0 to 6)
```



Example II – b) Step input - MATLAB

A system has the following transfer function:

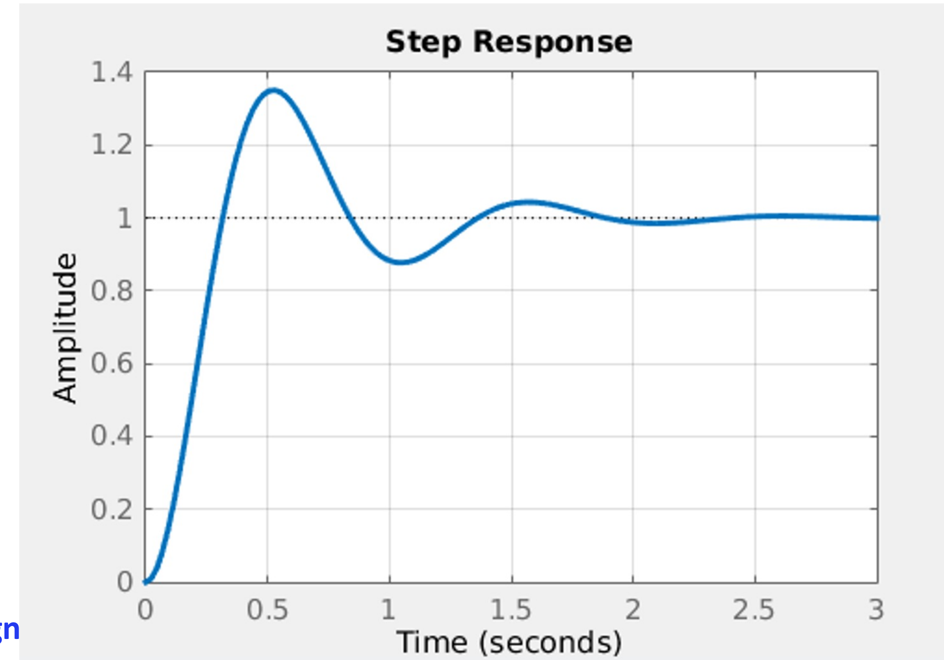
$$u(s) \longrightarrow \boxed{\frac{40}{s^2 + 4s + 40}} \longrightarrow y(s)$$

With a step of magnitude 1, what is the output as a function of time?

```
s = tf('s') % define s variable
G = 40/(s^2 + 4*s + 40) % define transfer
                          function
step(G, 0:0.001:6) % Time response to a step
                     of 1 (from 0 to 6)
grid
```

OR

```
G = tf([40],[1 4 40])
step(G)
grid
```



Example II - Step input - MATLAB Simulink

A system has the following transfer function:

$$u(s) \longrightarrow \boxed{\frac{40}{s^2 + 4s + 40}} \longrightarrow y(s)$$

With a step of magnitude 1, what is the output as a function of time?

Example II - Step input - MATLAB Simulink

A system has the following transfer function:

$$u(s) \longrightarrow \boxed{\frac{40}{s^2 + 4s + 40}} \longrightarrow y(s)$$

$$y(s) = \frac{40}{s^2 + 4s + 40} u(s)$$

$$\Leftrightarrow \ddot{y} = -40y - 4\dot{y} + 40u$$

With a step of magnitude 1, what is the output as a function of time?